

Bank Customer Churn Analysis

Exploring Behavioural Drivers and Building a Predictive Model for Retention

The Churn Crisis

Understanding the 20.37% leak and the cost of institutional silence

Dataset Foundations

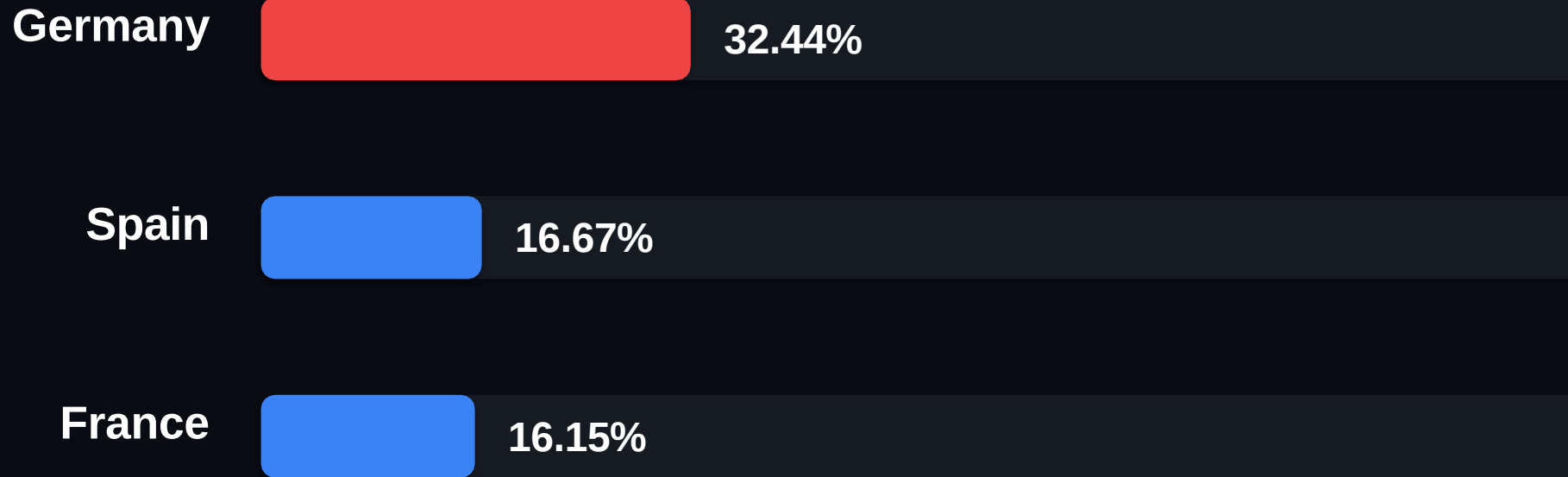
10,000 Records

Cleaned European banking portfolio encompassing 10k unique customer relationships across France, Germany, and Spain.

14 Critical Features

Comprehensive behavioral tracking: Credit scores, balance tiers, product counts, tenure, and active transactional markers.

Regional Churn Disparities



** Germany is the primary operational leak, experiencing nearly double the churn rate of neighboring branches despite representing only ~25% of the customer base.*

The Age Risk Factor

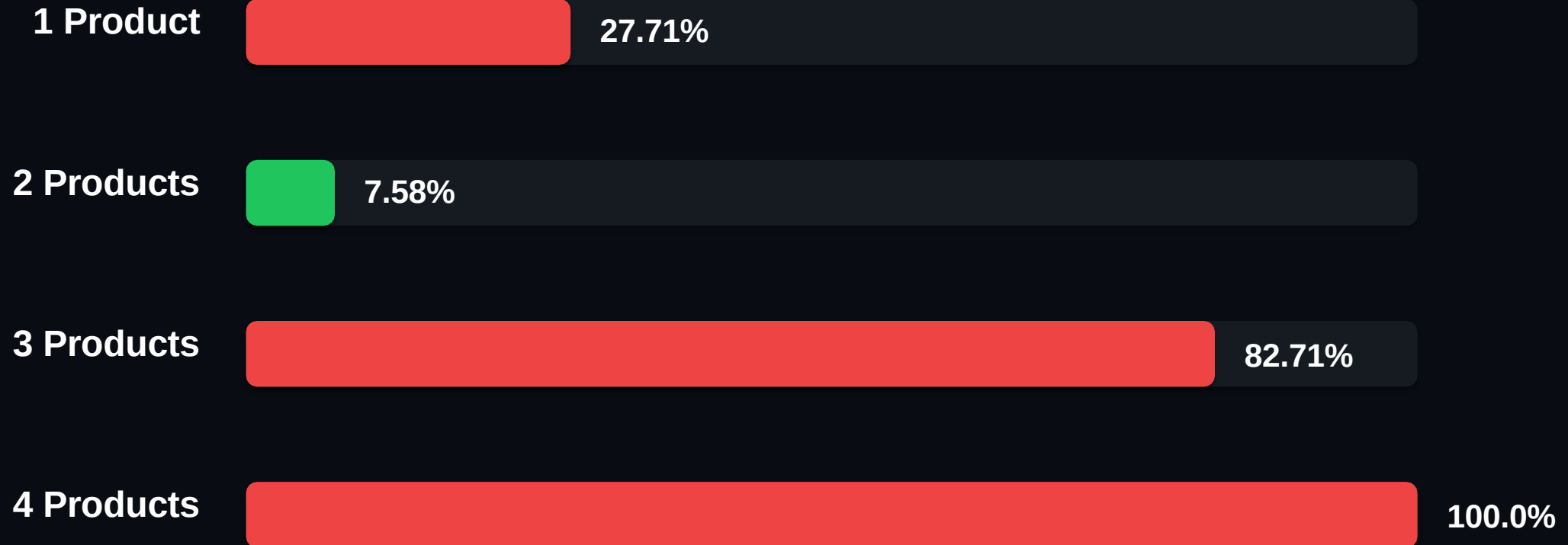
51.12%

Churn Rate in Older Adults

Structural Demographic Exposure

Customers aged 46–60 represent a critical emergency segment. While loyalty is high among Young Adults (7.52%), attrition spikes heavily among older adults (peaking at ~51%), before dropping significantly among senior accounts (24.78%).

The "Rule of 2" Paradox



Holding exactly two products serves as a massive loyalty anchor. Cross-selling beyond this threshold triggers catastrophic, near-total customer defection.

The Active Member Shield

Engagement as the Retention Anchor

- Maintaining an 'Active Member' status effectively slashes churn risk by roughly half across France, Germany, and Spain.
- This effect is profound in Germany, where moving a customer from a passive to an active member state drops their regional churn risk from an extreme 41.08% down to a manageable 23.72%.
- This retention benefit also remains remarkably consistent across age brackets.

Statistical Validation (Chi-Square)

Using level of significance (α) = 0.05

- H_0 : There is no relationship between Active Status/Credit Cards and Churn (they are independent)
- H_1 : There is a statistically significant relationship.

Customer Activity

p-value < 0.000001

Reject Null Hypothesis.

Active status is a mathematically proven anchor for customer retention.

Credit Card Hold

p-value = 0.492372

Fail to Reject Null.

Owning a credit card has no independent statistical connection with churn.

Core Takeaway

Don't waste budget pushing credit cards as a core retention tool.

Shift resources entirely to interactive membership and usage campaigns.

Predictive Machine Learning Model

86.05%

Overall Model Classification Accuracy (Random Forest Classifier)

Provides a robust, ready-to-deploy early warning system to flag high-risk accounts before they exit.

Model Operational Value Matrix

78% Precision

When the model flags a customer as likely to churn, approximately 78% of those flagged customers actually churned. This indicates that the model's alerts are relatively targeted, although approximately 22% of flagged customers would still be false positives.

40% Recall

The model identifies approximately 40% of customers who actually churned, meaning around 60% of churners are missed. This conservative behavior reduces false alarms but may limit the bank's ability to identify customers who could benefit from early retention interventions.

Strategic Actions

- ✓ **Cap Cross-Selling at the "Rule of 2":** Halt automated cross-selling beyond 2 products to avoid near-total customer defection. Audit existing accounts holding 3+ products to identify bottlenecks driving exit decisions.
- ✓ **Audit German Branch Disparities:** Audit regional policies and user experience to address Germany's 32.44% churn leak, prioritizing immediate retention interventions for the high-risk German female segment.
- ✓ **Deploy Older Adult Engagement:** Design specialized wealth-preservation perks for the 46–60 age bracket (51.12% baseline churn), prioritizing active account status to anchor long-term savings.
- ✓ **Pivot Strategy Toward Active Engagement:** Allocate capital toward retention programs that incentivize recurring account interaction and sustain long-term activity.

Questions

Thank you for your time.